

Setting Up Bacula, the Modular Backup Solution

Bacula is an open source, enterprise-level computer backup system for heterogeneous networks, which is distributed and highly customisable. It is designed to automate backup tasks that often require intervention from a systems administrator or computer operator. This article introduces readers to the basics of Bacula.



Let me begin with sharing a real-life example of my use of Bacula. One of our projects involved using virtual machines installed on our workstation. This allowed us to quickly spin up different environments that were needed for development and testing. However, we needed a robust backup solution that allowed us to save the information not only on our VMs but also on our other machines that were running different operating systems. VMs come with the ability to take snapshots of the current state and we experimented with this feature,

as well as exporting an image of the current state of the virtual machine as a backup solution. However, this proved time consuming and didn't provide a solution for our Windows and Linux hosts. So we searched for a more robust backup solution, which ultimately turned out to be Bacula.

Architecture

Bacula comprises the following five major components or services —director, console, client, storage and catalogue.

It is a completely modular backup solution, allowing us to scale quickly by moving the

individual components onto their own computers, or even hosting multiple computers – each running their own file, storage or database servers. Even with multiple database servers, or multiple file servers, we can still manage everything from one director instance. For example, we can add new file, database and storage servers to the director's configuration file, thus expanding the capabilities of the backup solution. However, the capabilities of Bacula are far too many to list here. In fact, this article aims to provide just enough information to get readers started.